

NATIONAL-LEVEL HACKATHON PROJECT DOCUMENTATION

Myra's Global Tech HACK-AI-THON 2026

TerraLoop

“Measure Waste. Close the Loop.”

A Python Desktop Application for Household Waste Tracking, Recycling Analysis & Sustainability Insight

Technology Stack	Python, Tkinter, Pillow (PIL)
Project Type	Object-Oriented desktop application
Theme	Python Fundamentals + OOP + Sustainable Development Goals
SDG Alignment	SDG 11 – Sustainable Cities and Communities SDG 12 – Responsible Consumption and Production SDG 13 – Climate Action
Team Name	HackAIThon Group 3
Team Members	Satvik, Kanav, Nipun and Tanvi
GitHub Repository	https://github.com/satvik153/HackAlthon_Group3/tree/main

1. Executive Summary

TerraLoop is a Python desktop application that helps households record, understand, and act on the waste they generate. Built with Tkinter for the interface and structured around object-oriented design, the application lets a user log individual waste entries by category and weight, then turns that raw data into readable insights: total waste generated, recycling rate, the category responsible for the most waste, and a custom sustainability score. Household waste is generated every day, but it is rarely measured. Most people can estimate roughly how much they throw away, but few can say with confidence which category — plastic, organic, paper, glass, metal, or e-waste — dominates their footprint, or whether their recycling habits are actually improving over time. TerraLoop addresses this gap by giving users a simple, structured way to record waste as it happens and see the resulting patterns immediately.

Beyond tracking, TerraLoop analyzes the recorded data to highlight the household's “waste hotspot,” generates rule-based recommendations targeted at that hotspot, and includes a What-If Simulator that lets users project the effect of reducing a category by a chosen percentage. Together, these features turn a static log of numbers into a decision-support tool, aligning directly with the hackathon theme of applying Python fundamentals and OOP to a Sustainable Development Goal problem.

2. Problem Statement

Households generate multiple types of waste every day — plastic packaging, food scraps, paper, glass, metal, and occasionally electronic waste — but almost none of this is tracked in any structured way. As a result, several basic questions usually go unanswered:

- How much total waste does the house hold actually generate over a week or a month?
- Which category of waste contributes the most to that total?
- What proportion of the waste generated is actually being recycled?
- Is recycling behaviour improving, staying flat, or getting worse over time?

Without this information, waste-management decisions at the household level are based on assumptions and habit rather than on measurable evidence. A household might assume it recycles “most” of its waste, or that plastic is a minor issue, when the opposite could be true. This lack of visibility makes it difficult to target effort where it would have the most impact — whether that effort comes from an individual family, a housing society, or a local awareness campaign.

A simple digital tracking and analysis system closes this gap. By recording each waste entry with a category, weight, date, and recycling status, even a small amount of structured data is enough to calculate meaningful metrics and surface patterns that would otherwise stay invisible. This is the core problem TerraLoop is built to solve.

3. Existing Gap & Proposed Solution

3.1 Existing Gap / Challenges

- Households have no simple tool to record waste at the point of disposal, so data is never captured in the first place.
- Without recorded data, identifying a waste “hotspot” —the category generating the most waste —is guesswork.
- Recycling performance is rarely measured, so it is hard to tell whether current habits are actually effective.
- Even when a household is aware of a problem area, there is no structured way to receive actionable, specific suggestions.
- It is difficult to judge, in advance, how much a behavior change (such as reducing plastic use) would actually improve outcomes.

3.2 Proposed Solution

TerraLoop provides a lightweight desktop application where a household can log every waste item as it is generated, categorising it and marking whether it was recycled. The application's analysis layer then converts these individual entries into aggregate statistics — total waste, recycling rate, category-wise percentages, and the current hotspot — and its recommendation engine translates those statistics into specific, rule-based suggestions. A What-If Simulator lets the user test hypothetical improvements before committing to them in real life, making the impact of a habit change tangible rather than abstract.

3.3 Objectives

- Provide a simple digital interface to record household waste by category, weight, date, and recycling status.
- Allow users to add, edit, delete, and search waste records, and filter them by time period.
- Analyze recorded data to calculate total waste, recycled waste, and recycling rate.
- Identify the waste category that contributes the most — the household's waste hotspot.
- Generate rule-based, actionable recommendations targeted at the household's specific waste pattern.
- Enable users to simulate the effect of reducing a waste category by a chosen percentage.
- Encourage more consistent, evidence-based waste and recycling habits through visibility and feedback.

4. System Overview & Architecture

TerraLoop follows a linear, layered flow: raw user input becomes structured records, records are aggregated into insights, and insights are translated into guidance the user can act on. Each stage is owned by a dedicated class, keeping data handling, analysis, and presentation logic separate.

4.1 Data & Analysis Flow

Stage	Description
User Input	Waste is entered through the Add Waste screen (category, weight, date, recycled flag, note).
Waste Records	Each entry is stored as a WasteRecord object held in memory for the session.
Waste Manager	Adds, edits, deletes, searches, and filters records by category or time period.
Waste Analyzer	Computes totals, recycling rate, category percentages, and the waste hotspot.
Insights	Waste Hotspot · Recycling Rate · Category Analysis · Sustainability Score
Recommendation	Applies rule-based conditions to the insights to produce targeted suggestions.
Engine User Actions	The user reviews recommendations, runs simulations, and adjusts real habits.

4.2 What-If Simulator (Overview)

The simulator sits alongside the main analysis flow rather than inside it. It takes the current set of records, applies a hypothetical reduction to one category, and re-runs the same calculations used elsewhere in the application (total waste, category percentages, recycling rate) on the adjusted figures. The result is shown next to the current, real values so the user can compare “as-is” against “what-if” directly. Section 8 covers this in detail.

4.3 Core Classes & Responsibilities

Class	Responsibility
Household	Stores household-level information used to contextualise the records.
Waste Record	Represents a single waste entry — category, weight, date, recycled status, note.
Waste Manager	Adds, updates, deletes, searches and filters the collection of WasteRecord objects.
Waste Analyzer	Performs calculations: totals, recycling rate, category percentages, hotspot detection.
Recommendation Engine	Applies rule-based conditions to analyzer output to generate recommendations.
Sustainability Simulator	Builds a modified “what-if” scenario and compares it with current results.
TerraLoop App	Builds the Tkinter interface and wires the other classes together.

5. Object-Oriented Programming Implementation

TerraLoop's structure was deliberately built around OOP fundamentals, both because the hackathon theme calls for it and because it keeps the codebase easy to extend — new waste categories, new metrics, or new recommendation rules can be added without touching unrelated parts of the program.

5.1 Classes and Objects

Every real-world entity in the problem domain — a household, a single waste entry, the collection of records, the analysis logic, the recommendation logic, and the simulator — is modelled as its own class. At runtime, each waste entry the user logs becomes a WasteRecord object, and the growing list of these objects is what the WasteManager, WasteAnalyzer, and RecommendationEngine all operate on.

5.2 Encapsulation

Each class groups the data and behaviour that belong together. A WasteRecord keeps its own category, weight, date, and recycled flag together with any validation needed when that data is created or edited, instead of scattering these fields across the application. Similarly, WasteManager is the only class that directly manipulates the internal list of records — other classes ask it for data rather than reaching into that list themselves.

5.3 Abstraction

The interface layer never needs to know how a metric is calculated — only that calling the appropriate WasteAnalyzer method returns a ready-to-display result. For example, the Dashboard simply requests the recycling rate and the current hotspot; the arithmetic, filtering, and comparison logic behind those numbers stays inside WasteAnalyzer. This separation means the calculation logic could be improved later without changing how the dashboard displays it.

5.4 Composition / Collaboration

TerraLoopApp does not implement waste logic itself; it composes the other classes and coordinates them. WasteManager supplies records to WasteAnalyzer, WasteAnalyzer's output is passed to RecommendationEngine to generate suggestions, and SustainabilitySimulator reuses WasteAnalyzer's own calculation methods on a modified copy of the data rather than duplicating that logic. This collaboration between focused classes, rather than one large class doing everything, is the core OOP contribution of the project.

5.5 Representative Methods

Method (illustrative)	Responsibility
WasteManager.add_record()	Validates and stores a new WasteRecord.
WasteManager.filter_by_period()	Returns records within a selected time window (e.g. Last 7 Days).
WasteAnalyzer.calculate_recycling_rate()	Computes recycled weight as a percentage of total weight.
WasteAnalyzer.get_hotspot()	Returns the category with the highest total recorded weight.
RecommendationEngine.generate_recommendations ()	Evaluates rules against analyzer output and returns suggestions.
SustainabilitySimulator.run_simulation()	Applies a reduction to one category and recalculates key

Method (illustrative)	Responsibility
	metrics.

A short snippet illustrating encapsulation – kept minimal, for concept demonstration rather than a full listing:

```
classWasteRecord:
    def __init__(self, category, weight, date, recycled, note=""):
        self.category = category
        self.weight = weight
        self.date = date
        self.recycled=recycled
        self.note = note
```

6. Core Algorithms & Calculations

TerraLoop's insights are all derived from simple, transparent arithmetic performed on the recorded WasteRecord objects. Keeping these calculations straightforward was a deliberate choice — every number shown to the user can be traced back to the raw records without a hidden model in between.

6.1 Total Waste

Total Waste = sum of the weight field across all recorded (or filtered) WasteRecord entries.

6.2 Recycled Waste

Recycled Waste = sum of the weight field for entries where the recycled flag is True.

6.3 Recycling Rate

Recycling Rate = $(\text{Recycled Waste} \div \text{Total Waste}) \times 100$

This single percentage gives an immediate sense of how much of the household's waste is being diverted from landfill, and is the headline metric shown on the Dashboard.

6.4 Category Percentage

Category Percentage = $(\text{Category Waste} \div \text{Total Waste}) \times 100$, calculated independently for each of the six categories: Plastic, Paper, Organic, Glass, Metal, and E-Waste.

6.5 Waste Hotspot

The category with the highest Category Percentage (equivalently, the highest total recorded weight) is flagged as the household's waste hotspot. This single indicator directs the user's attention to the one category where a change in habit would have the largest measurable effect, and it is the primary input to the Recommendation Engine.

6.6 Period Filtering

All of the calculations above can be run over a specific time window rather than the full record history. TerraLoop supports Last 7 Days, Last 30 Days, and All Time views, letting a user distinguish recent behaviour from their overall trend — for instance, a household could have a high all-time hotspot in Plastic but see that figure improving over the last 7 days.

These calculations are intentionally simple rather than statistical or predictive. This keeps the results fully explainable: a judge or user can manually verify any figure shown in the application by adding up the underlying records.

7. Recommendation Engine & Sustainability Score

7.1 Rule-Based Recommendation Engine

TerraLoop's Recommendation Engine uses a set of predefined conditional rules — evaluated against the output of Waste Analyzer — to decide which suggestions to surface. It does not use machine learning, training data, or predictive modelling of any kind.

- If a category represents a high proportion of total waste, the engine recommends reducing or better managing that specific category.
- If the overall recycling rate is low, the engine recommends improving segregation and recycling practices.
- If e-waste is present in the records, the engine adds a responsible e-waste disposal recommendation.

This is a rule-based recommendation system, not a machine-learning or AI model. Each recommendation is generated because a specific, checkable condition in the recorded data was met — for example, a category exceeding a percentage threshold, as shown in the Recommendations screen (Section 9).

7.2 Sustainability Score

TerraLoop calculates a custom, project-defined Sustainability Score (0–100) intended to give users a single, easy-to-read indicator of how their recorded waste behaviour is trending. The implementation combines factors already computed elsewhere in the application, including the recycling rate, how evenly waste is distributed across categories, and whether e-waste is being handled responsibly, into one weighted figure.

The Sustainability Score is a custom prototype metric defined for this project. It is not an officially standardized environmental index, has not been scientifically validated, and should be read as a directional, easy-to-understand indicator rather than an authoritative measurement.

Future versions of TerraLoop could replace or supplement this prototype score with validated environmental indicators and real-world reference datasets, once such data sources are integrated — see Section 11 for a full discussion of future scope.

8. What-If Sustainability Simulator

The What-If Simulator lets a user ask a concrete question — “what would happen if I reduced a particular waste category by a certain percentage?” — and see the answer immediately, calculated from their own recorded data rather than a generic estimate.

8.1 Workflow

- 1. The user selects a waste category from the current records (e.g. Plastic).
- 2. The user specifies a reduction percentage using the on-screen control (e.g. 30%).
- 3. TerraLoop creates a simulated scenario by reducing that category's recorded weight by the chosen percentage.
- 4. The application recalculates total waste, category percentages, recycling rate, and hotspot on the simulated data, reusing the same WasteAnalyzer logic described in Section 6.
- 5. The current (real) results and the simulated results are shown for comparison.

8.2 Why This Feature Is Useful

- Makes sustainability improvement measurable instead of aspirational—the user sees a projected number, not just advice.
- Helps users understand how much a specific change would actually move their overall recycling rate or hotspot.
- Turns the statistics generated in Sections 6 and 7 into a concrete, testable scenario.
- Encourages experimentation:a usercan try several categories or percentages before deciding where to focus effort.

8.3 Simplified Flow

Select Category → Set Reduction % → Run Simulation → Recalculate Metrics → Compare Current vs Simulated

9. Application Features & Screenshots

TerraLoop's interface is organised into seven sections accessible from a persistent side navigation: Dashboard, Add Waste, Records, Hotspot & Analysis, Recommendations, What-If Simulator, and Report. The screenshots below are taken directly from the working application.

9.1 Dashboard

The Dashboard is the application's landing screen. It summarises the household's current state in four cards — total waste, recycling rate, sustainability score, and number of records — and calls out the current waste hotspot alongside a short mission statement and the SDGs the project supports.

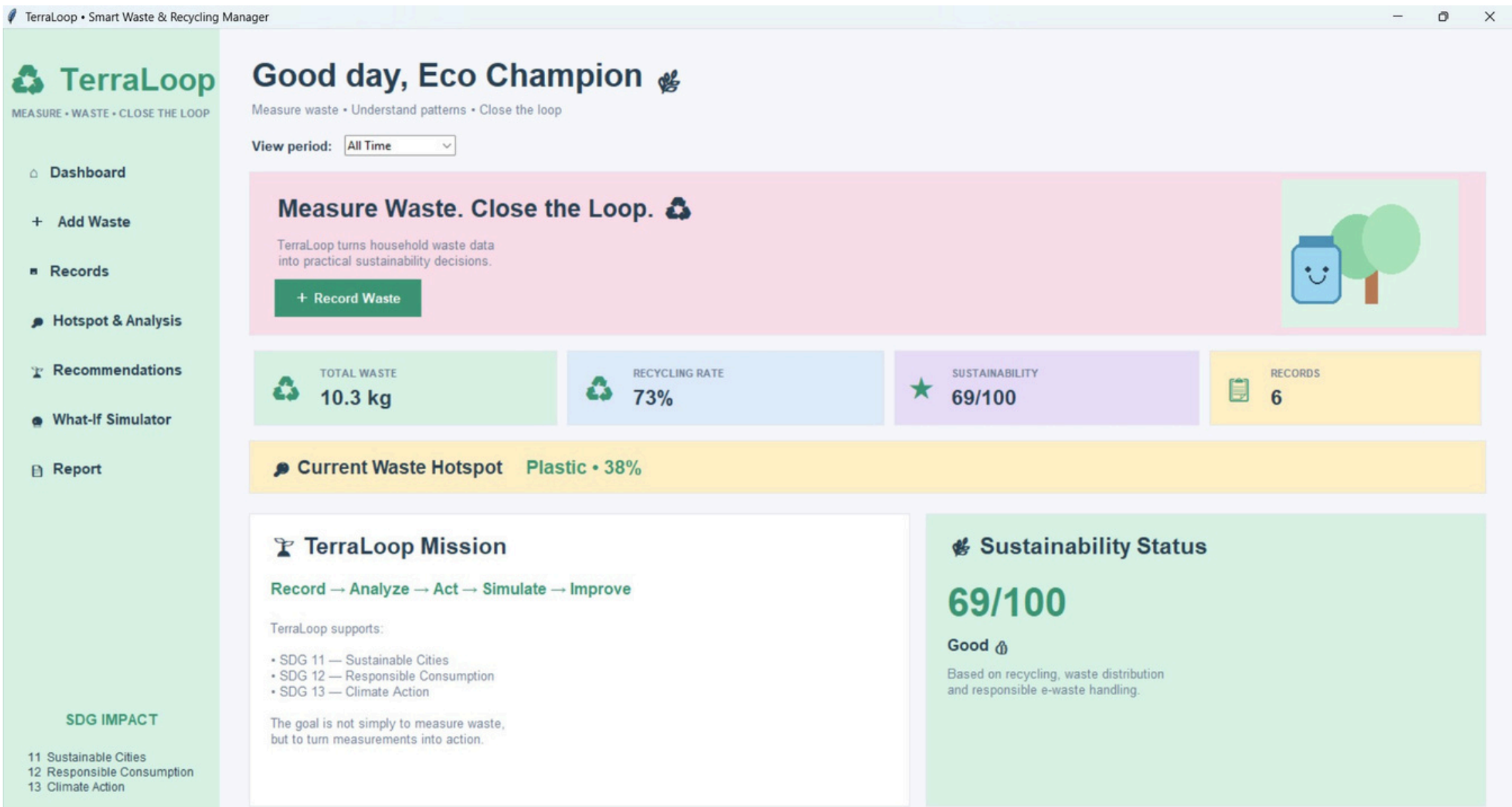


Figure 1. Dashboard — summary metrics, current hotspot, and SDG alignment at a glance.

9.2 Add Waste

The Add Waste screen is where new entries are logged: category, weight in kilograms, date, a recycled checkbox, and an optional note. This is the single data-entry point that feeds every downstream calculation.

The screenshot shows the 'Add Waste Record' form. It has a 'Category' dropdown menu with 'Plastic' selected. Below it is a 'Weight (kg)' input field. Then a 'Date (YYYY-MM-DD)' input field with '2026-09-14' entered. A 'Recycled?' checkbox is present. Below that is a 'Note' input field. At the bottom is a green 'SAVE RECORD' button.

Figure 2. Add Waste — category, weight, date, and recycled status entry.

The screenshot shows the 'Waste Records' table. It has a search bar and a 'Category' dropdown menu set to 'All'. The table has columns: ID, Date, Category, Weight, Status, and Note. Below the table are 'Edit' and 'Delete' buttons.

ID	Date	Category	Weight	Status	Note
6	2026-09-11	Plastic	2.1 kg	Not Recycled	Mixed plastic
5	2026-09-09	E-Waste	0.7 kg	Not Recycled	Old cables
4	2026-09-07	Glass	1.1 kg	Recycled	Glass jars
3	2026-09-05	Paper	1.4 kg	Recycled	Newspapers
2	2026-09-03	Organic	3.2 kg	Recycled	Kitchen waste
1	2026-09-01	Plastic	1.8 kg	Recycled	Bottles and packaging

Figure 3. Records — search, category filter, edit and delete.

9.3 Records

The Records screen lists every logged entry with its ID, date, category, weight, recycling status, and note. Users can search by keyword, filter by category, and select a record to edit or delete, giving full control over the data behind the analysis.

9.4 Hotspot & Analysis

This screen breaks total waste down by category with a percentage bar for each, and states the main hotspot explicitly — the category responsible for the largest share of recorded waste, as calculated in Section 6.5.



Figure 4. Hotspot & Analysis — category breakdown with the main hotspot called out.

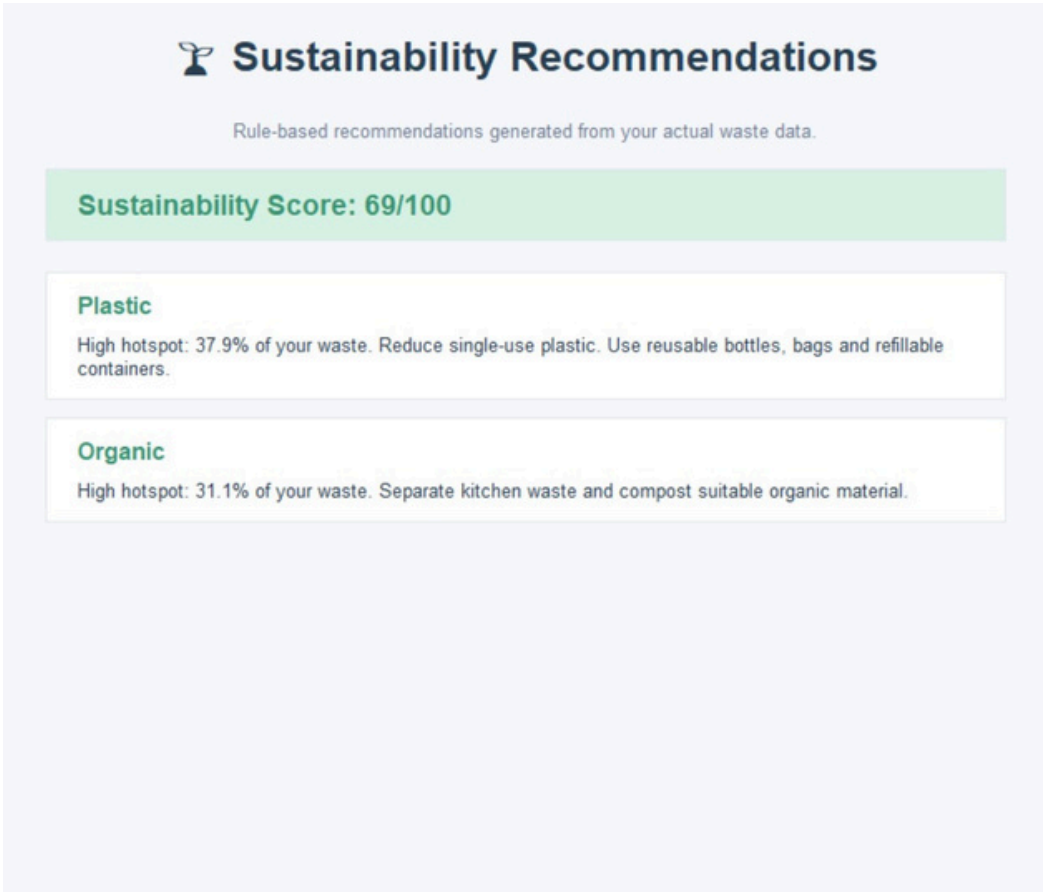


Figure 5. Recommendations — rule-based suggestions with the sustainability score.

9.6 What-If Simulator

The simulator screen lets the user choose a category and a reduction percentage via a slider, then run the simulation to see projected results next to the current figures, as described in Section 8.

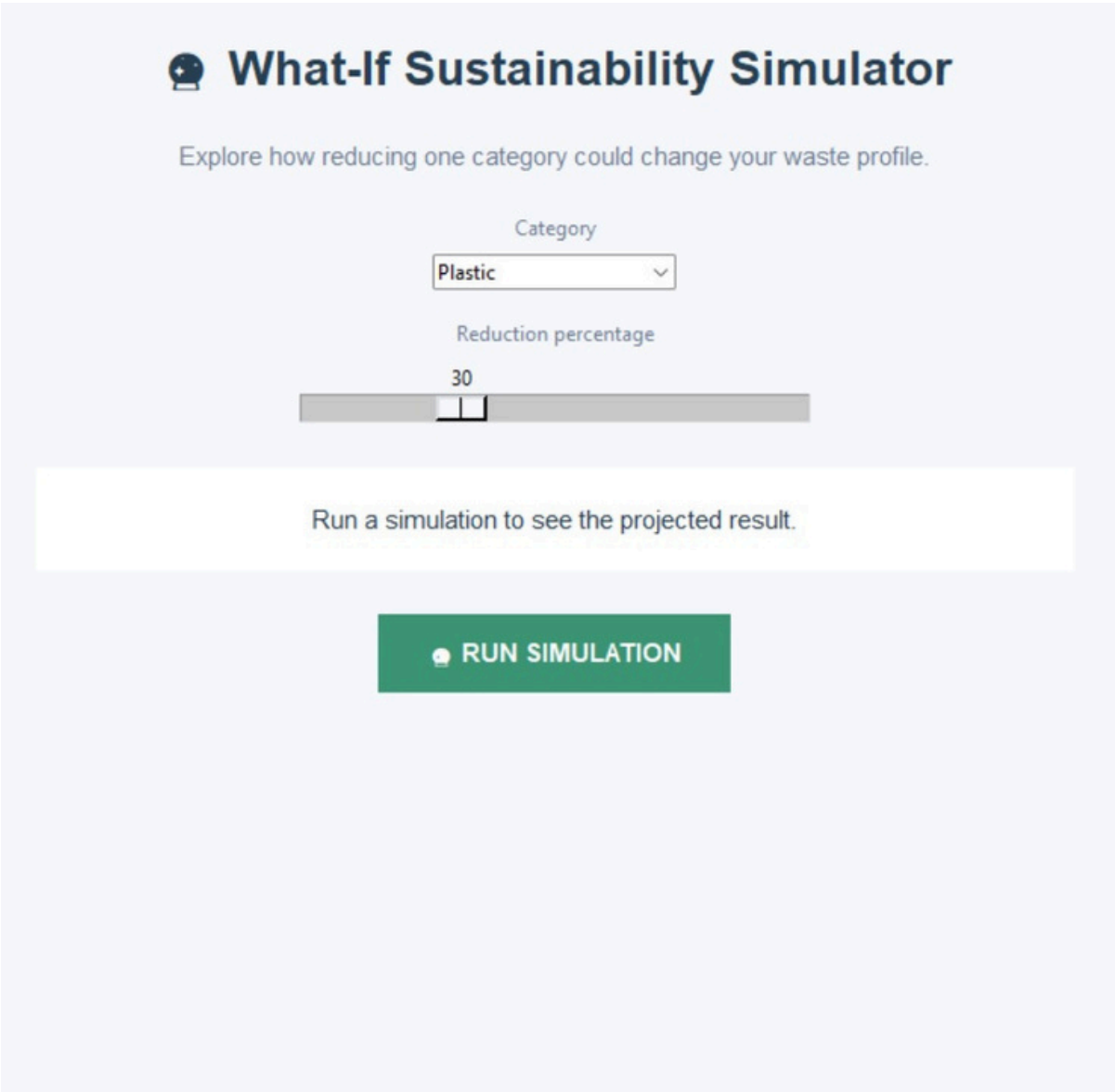


Figure 6. What-If Simulator — category, reduction percentage, and simulation trigger.

10. Testing & Validation

TerraLoop was validated through manual functional testing using meaningful, hand-entered sample data covering all six categories, a mix of recycled and non-recycled entries, and a range of dates and weights. Each core feature was exercised directly through the interface and checked against the expected result. No automated unit-test suite has been built for this prototype; testing to date has been manual and feature-by-feature.

Test	Input / Action	Expected Result
Add valid record	Valid category + weight entered	Record added to list
Invalid weight	Negative or non-numeric value entered	Validation message shown
Future date	Date later than today entered	Input rejected
Edit record	Change an existing record's fields	Record updated in place
Delete record	Delete a selected record	Confirmation prompt, then deletion
Search	Search by category or note keyword	Matching records displayed
Category filter	Select a category from the filter	Only relevant records shown
Calculation accuracy	Run totals against sample data	Statistics match manual calculation
Recommmendation trigger	Sample data with low recycling rate	Relevant recommendation generated
Simulation	Reduce a selected category by a set %	Scenario recalculated and displayed
Report generation	Generate summary report	Summary produced with current figures

11. Sample Data, Limitations & Future Scope

11.1 Sample Data

The application was demonstrated using a small set of meaningful sample records spanning all six waste categories, a mix of recycled and non-recycled entries, several different dates within a two-week window, and a realistic range of weights (roughly 0.7 kg to 3.2 kg per entry). This was enough sample data to produce non-trivial totals, a clear hotspot, and a recycling rate meaningfully below 100%, so that every analysis and recommendation feature could be demonstrated with genuine output rather than placeholder values.

11.2 Limitations

- Data is currently held in memory during runtime and is not saved permanently once the application is closed.
- The Sustainability Score is a custom prototype metric, not a scientifically validated or externally standardized index.
- The Recommendation Engine is rule-based; it does not use machine learning or predictive modelling.
- There is no connection to an external, real-world waste or recycling database.
- TerraLoop is a single-user desktop application rather than a cloud-connected or multi-device system.

11.3 Future Scope

Implemented features are described above; the following are realistic directions for future versions and are not part of the current prototype:

- Persistent local storage(e.g. a file-based or SQL database)so records survive between sessions.
- User accounts to support multiple house holds or family members on one installation.
- Cloud synchronization for access across devices.
- A mobile or web version of the application.
- Integration with real-world waste and recycling data sets for benchmarking.
- More advanced analytics, such as trend charts over longer time periods.
- A more detailed, multi-factor sustainability metric, potentially aligned with external standards.
- Optional machine-learning-based prediction(e.g. forecasting future waste volume)as a clearly separate, opt-in feature in later versions.

12. SDG Alignment, Impact & Conclusion

12.1 SDG Alignment

SDG 11 — Sustainable Cities and Communities

By giving households a way to measure and understand their own waste generation, TerraLoop supports more sustainable local waste management practices — a foundational element of sustainable communities.

SDG 12 — Responsible Consumption and Production

The recycling rate, category breakdown, and hotspot analysis directly encourage more responsible consumption habits by making the consequences of those habits visible and specific.

SDG 13 — Climate Action

Reducing and better managing household waste, particularly categories like plastic that carry a significant production and disposal footprint, contributes in a small but real way to broader climate action goals.

12.2 Project Impact

At the scale of an individual household, TerraLoop's realistic impact is behavioural: it replaces vague assumptions about waste with specific, checkable numbers, and turns general advice (“recycle more”) into a targeted suggestion tied to that household's own hotspot. If adopted more broadly — for instance across a housing society or a school awareness programme — the same tool could help a community identify shared patterns and prioritise where recycling infrastructure or awareness efforts would matter most. These are presented as reasonable, modest outcomes appropriate to a prototype, not as measured results.

12.3 Team Contributions

Members	Contributions
Kanav	Made the code
Satvik	Prepared documentation and github repository
Nipun	Made the presentation
Tanvi	Made and explained the video

12.4 Conclusion

TerraLoop brings together Python, object-oriented design, and straightforward data analysis to address a genuine, everyday gap: households generate waste continuously but rarely measure it. By structuring the problem around focused classes — for records, management, analysis, recommendations, and simulation — the project demonstrates practical OOP fundamentals while producing a tool that is immediately usable. Its rule-based recommendations and What-If Simulator turn raw numbers into decisions a household can actually act on, and its honestly-scoped limitations leave clear, realistic room for growth. As a prototype built for Myra's Global Tech HACK-AI-THON 2026, TerraLoop shows how a modest, well-structured piece of software can make sustainability measurable at the household level.